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Resolution Effects in the Measurement of Phonons in Sodium Metal*

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In the course of making a detailed study of the phonons in Na metal using a triple-axis neutron spectrometer we have observed anomalous, asymmetric, and, in some cases, extraordinarily broad or split resonances. In this paper we account for these effects quantitatively by performing a complete resolution analysis. This analysis includes fully the *combined* effects of vertical resolution, a highly anisotropic dispersion surface, the variation of the polarization vectors and the temperature factor inside the resolution volume, and the sample mosaic spread. The application of this calculation to the measurement of phonons in other materials is straightforward.

I. INTRODUCTION

In this paper we describe results which were obtained from a detailed study of phonons in sodium metal. During our experiments, which were performed with a triple-axis neutron spectrometer, we observed anomalous, asymmetric, and in some cases, extraordinarily broad or split resonances in the energy spectra of neutrons scattered from the sample. Additional resonances, corresponding to neutron-scattering vectors $\vec{Q}$ in the $\langle 100\rangle$ direction, have been observed in several alkali metals.[1-3] These resonances appear to indicate the existence of an extra, longitudinally polarized, phonon mode propagating in the $\langle 100\rangle$ direction, whose frequencies lie below that of the transverse branch. In an effort to decide whether these anomalous, extra resonances are an instrumental effect or of physical importance in the alkali metals, we have investigated them in some detail in Na. Resonances corresponding to phonons in the $\langle 111\rangle$ direction of Na have been found to be much broader (with an indication of splitting) than a simple resolution calculation would predict, and they are generally asymmetric. In addition, some of the low $\vec{q}$ longitudinal phonons are asymmetric.

The purpose of this paper is to present a detailed resolution analysis in an effort to account for these effects quantitatively. Since a calculation of this type is sufficiently difficult (without the use of an excessive amount of computer time) and its application to the analysis of phonon resonances in other materials is straightforward, a complete description of the analysis and computation is given.

The resolution function which we use is that given by Cooper and Nathans,[4] with certain modifications which account for the various beam-path configurations and the effects of the mosaic spread of the sample. It is our intent to emphasize the practical difficulties involved in obtaining a reliable set of instrumental parameters and the approximations involved in a resolution calculation of this kind. The importance of the application of an accurate and reliable resolution calculation in the neutron measurement of phonon lifetimes, line shapes, and polarization vectors cannot be overestimated.

II. FORMULATION OF THE RESOLUTION PROBLEM

A. Preliminary Comments

In recent years a number of papers on the resolution of triple-axis neutron spectrometers have been published.[4-6] A review of these articles has recently been given by Møller and Nielsen[7] who also present certain extensions and simplifying ideas. However, it appears that there is no published work which fully includes the combined effects of vertical resolution, a highly anisotropic dispersion surface, the variation of polarization vectors inside the resolution volume, and the sample mosaic spread which is necessary in analyzing our data on Na. The results of a calculation which is similar to the one presented here have recently been published by Samuelsen *et al.* for the case of magnons in Cr_2O_3.[8] Since it has been necessary to make certain important modifications to the Cooper-Nathans resolution function, we will outline in this section the derivation of the expression for the counting rate which we have used.

If the spectrometer is set to observe a scattering process involving energy and momentum transfers $\hbar\omega_0$ and $\hbar\vec{Q}_0$, respectively, then the conservation rules for the nominal incident $\vec{k}_1^0$ and scattered $\vec{k}_2^0$ neutron wave vectors are

$$\vec{k}_2^0 - \vec{k}_1^0 = \vec{Q}_0 \tag{1}$$

and

$$(\hbar^2/2m_n)[(k_1^0)^2 - (k_2^0)^2] = \hbar\omega_0, \tag{2}$$

where we take ω_0 as positive for neutron energy loss. However, because of the finite collimation and mosaic spreads of the monochromating and analyzing crystals, scattered intensity may be observed from points in ω, $\vec{Q}$ space adjacent to ω_0, $\vec{Q}_0$. The resolution function of the spectrometer $R(\Delta\vec{Q}, \Delta\omega)$ is defined as the probability of detecting a neutron as a function of $\Delta\vec{Q}(=\vec{Q}-\vec{Q}_0)$ and $\Delta\omega(=\omega-\omega_0)$ when the instrument is set to measure the scattering at the point ω_0, $\vec{Q}_0$. It is *not* a distribution function as has often been mistakenly stated. It is simply a number dependent on $\Delta\vec{Q}$ and $\Delta\omega$.

B. Geometrical Interpretation of $R(\Delta\vec{Q}, \Delta\omega)$

The triple-axis spectrometer at Studsvik[9] on which these measurements were made consists of a double-crystal (copper 220) monochromator, a collimator placed between the monochromator and the sample, and an analyzing system using another copper crystal and a collimator placed between the sample and the analyzer. The primary collimation and the detector collimation are relaxed to the point where they do not contribute to the resolution. The spectrometer was generally operated in the constant-$\vec{Q}$ mode with fixed final energy.

The monochromating system (crystal and collimator), in effect, transfers neutrons from the volume V_0 in $\vec{k}$ space to the volume V_1, as shown in Fig. 1(a) with a probability dictated by the efficiency of the monochromating crystal(s) and the transmission of the incident collimator. In our case, with a double-crystal monochromator, those neutrons in V_1 are initially transferred to V_0 by the first crystal and then back to V_1 by the second crystal with probabilities governed by the efficiencies of the two crystals. The distribution of neutrons in V_1 which are incident on the sample is determined by the mosaic distribution of the monochromator (in our case, the product of the mosaic distributions of the two crystals) which gives the width of the distribution in the $\vec{\Delta}_1$ direction, and the transmission of the incident collimator which limits the distribution in the $\vec{\Delta}_2$ and $\vec{\Delta}_3$ (vertical) directions. Thus, the number of neutrons/cm^2/sec per unit volume in $\vec{k}$ space incident on the sample is

$$\Phi(\vec{k}_1) = \Phi_0 P_M(\vec{\Delta}), \tag{3}$$

where $\vec{k}_1$ deviates from the mean $\vec{k}_1^0$ by the vector $\vec{\Delta}$, that is,

$$\vec{k}_1 - \vec{k}_1^0 = \vec{\Delta} = \vec{\Delta}_1 + \vec{\Delta}_2 + \vec{\Delta}_3 . \tag{4}$$

Φ_0, which is the number of neutrons/cm^2/sec per unit volume in $\vec{k}$ space incident on the monochromator, is assumed to vary slowly over the small volume V_0 in comparison to the rapid variation of $P_M(\vec{\Delta})$ over the volume V_1. $P_M(\vec{\Delta})$ may be written as the product of three functions $P_M^{(1)}(\vec{\Delta}_1)$, $P_M^{(2)}(\vec{\Delta}_2)$, and $P_M^{(3)}(\vec{\Delta}_3)$ and is dimensionless.

The analyzing system accepts scattered neutrons in the volume V_2 as shown in Fig. 1(b) with a probability determined by the analyzing collimator transmission and the reflectivity of the analyzer, that is, by a function $P_A(\vec{\delta})$, which in analogy with $P_M(\vec{\Delta})$, is separable into three functions of $\vec{\delta}_1$, $\vec{\delta}_2$, and $\vec{\delta}_3$. Thus, the counting rate at a given setting of the spectrometer is

$$N = \Phi_0 \int P_M(\vec{\Delta}) \frac{d^3\sigma}{dk_2^3} P_A(\vec{\delta})\, d^3k_1\, d^3k_2 . \tag{5}$$

$d^3\sigma/dk_2^3$ is the cross section of the sample for scattering neutrons of wave vector $\vec{k}_1$ into d^3k_2 at $\vec{k}_2$. It is related to the usual expression for the inelastic scattering cross section by

$$\frac{d^3\sigma}{dk_2^3} = \frac{d^2\sigma}{d\Omega d\omega}\frac{\hbar}{m_n k_2}$$

$$= n_0 V \bar{b}^2 \frac{\hbar^2}{2m_n}\frac{1}{k_1}\frac{n_Q+1}{M\omega_Q}(\vec{Q}\cdot\vec{\epsilon})^2 e^{-2W}\delta(\omega-\omega_Q), \tag{6}$$

where the second expression is for one-phonon creation in a crystal with 1 atom/unit cell, e.g., sodium. n_0 is the atom density, V is the volume of the sample irradiated, $\bar{b}$ is the mean scattering length per atom, n_Q is the Bose population factor, M is the atomic mass, and $\vec{\epsilon}$ is the polarization vector.[10]

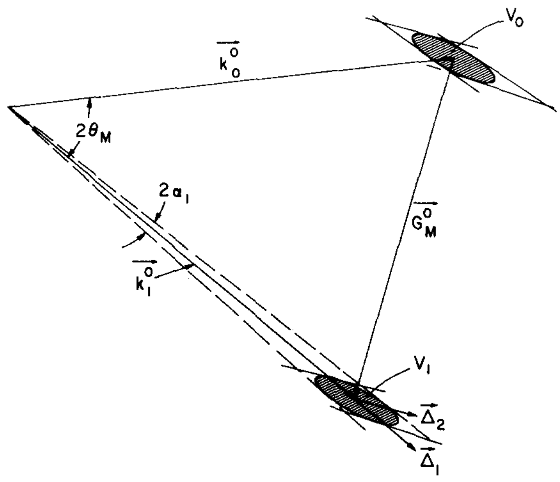
One would like to rewrite the integral in (5) as an integration of the product of a resolution function (dependent only on $\Delta\vec{Q}$ and $\Delta\omega$) and the cross section (as done by Cooper and Nathans). This can be done approximately. Holding $\Delta\vec{Q}$ and $\Delta\omega$ constant places four constraints on the integral, and the integration over the two remaining variables yields the resolution function. The constraints are given by

$$\Delta\vec{Q} = \vec{\delta} - \vec{\Delta} \tag{7}$$

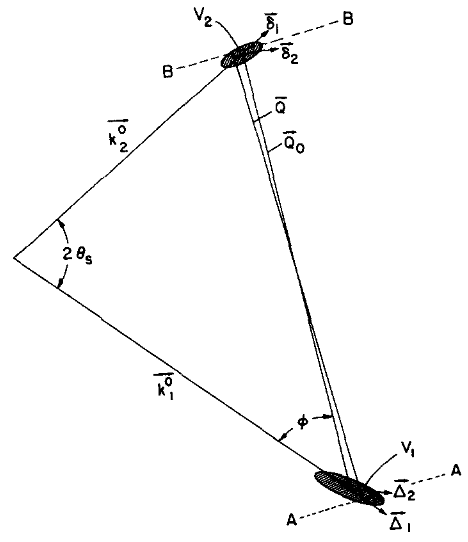
and

$$\Delta\omega = (\hbar/m_n)(\vec{k}_1^0\cdot\vec{\Delta} - \vec{k}_2^0\cdot\vec{\delta}). \tag{8}$$

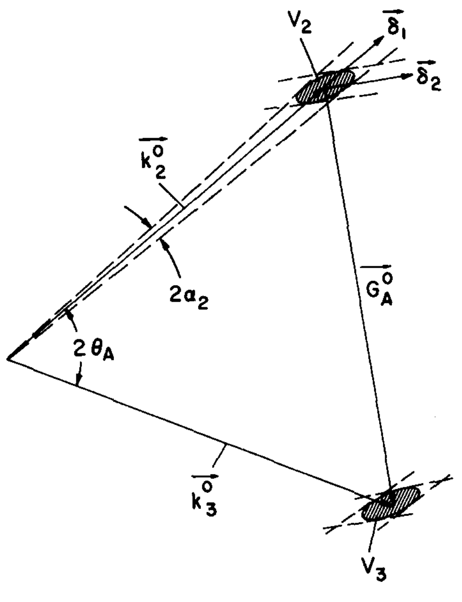
For a given $\Delta\vec{Q}$ and $\Delta\omega$, the tips of the vectors $\vec{k}_1$



(a)



(c)



(b)

FIG. 1. (a) The distribution of neutrons in $\vec{k}$ space incident on the sample is shown here. This distribution in the region V_1 is spread out in the $\vec{\Delta}_1$ direction due to the mosaic spread of the monochromator, and in the $\vec{\Delta}_2$ direction due to the finite collimation on the beam incident on the sample. These directions are chosen so that the distribution of incident neutrons can be separated into two functions depending on mosaic spread and collimation independently. (b) Scattered neutrons of wave vector $\vec{k}_2$ adjacent to $\vec{k}_2{}^0$ in the region V_2 of $\vec{k}$ space are accepted by the analyzer system. This "acceptance" function is spread out in the direction $\vec{\delta}_1$ due to the analyzer mosaic spread, and in the direction $\vec{\delta}_2$ due to the finite collimation on the scattered beam. (c) Incident neutrons of wave vector $\vec{k}_1$ distributed according to the function $P_M(\vec{\Delta})$ in the region V_1 of $\vec{k}$ space which are scattered by the sample into the volume V_2 are accepted by the analyzing system and counted with an efficiency determined by the function $P_A(\vec{\delta})$. For a given scattering vector $\vec{Q}=\vec{Q}_0+\Delta\vec{Q}$ and frequency $\omega=\omega_0+\Delta\omega$, the locus of allowed incident ($\vec{k}_1$) and scattered ($\vec{k}_2$) neutron wave vectors is restricted to lie on the planes A-A and B-B, respectively.

and $\vec{k}_2$ are restricted to lie on planes A-A and B-B, respectively, in Fig. 1(c). These planes are perpendicular to $\vec{Q}=\vec{k}_2-\vec{k}_1$, and there is a one-to-one correspondence between points (Δ_y, Δ_z) in the plane A-A and points (δ_y, δ_z) in the plane B-B. Thus (Δ_y, Δ_z) and (δ_y, δ_z) are the ends of the vector $\vec{Q}$. Accordingly we may write (5) as

$$N=\Phi_0\iint \frac{m_n}{\hbar Q}\frac{d^3\sigma}{dk_2^3}\, d^3(\Delta\vec{Q})\, d(\Delta\omega)\iint P_M(\vec{\Delta})P_A(\vec{\delta})d\Delta_y\, d\Delta_z \tag{9}$$

or

$$N=\Phi_0\frac{m_n}{\hbar}\iint \frac{1}{Q}\frac{d^3\sigma}{dk_2^3}\, R(\Delta\vec{Q}, \Delta\omega)\, d^3(\Delta\vec{Q})\, d(\Delta\omega), \tag{10}$$

where the resolution function is

$$R(\Delta\omega, \Delta\vec{Q})=\int\int P_A(\vec{\delta})P_M(\vec{\Delta})\, d\Delta_y\, d\Delta_z . \tag{11}$$

The factor $m_n/\hbar Q$, which is the Jacobian obtained in transforming the variables of Eq. (5) to those of Eq. (9), has generally been omitted,[4] but it is important to include its variation when the intensities of scattering from phonons of various wave vectors are to be compared. The approximation made in the above analysis consists of assuming that the $1/k_1$ factor in the cross section does not vary with Δ_y and Δ_z and in removing this factor from the integral expression for R. This approximation should, however, be reasonable since $|\Delta|/k_1$ is usually very small and thus $1/k_1$ can be replaced by $1/k_1^0$ and removed from the integral of Eq. (11). Naturally, the variation of $1/k_1^0$ as a function of spectrometer setting during a constant-$\vec{Q}$ scan must be taken into account. It will be noted that R has the dimensions length^{-2} as defined by Eq. (11).

If we follow the Cooper-Nathans formulation and approximate the transmissions of the collimators and the reflectivities of the monochromator and analyzer by Gaussian functions, the resolution function becomes

$$R = R_0 \exp(-\tfrac{1}{2}\textstyle\sum_{ij} M_{ij} X_i X_j), \tag{12}$$

where $(X_1,\ X_2,\ X_3,\ X_4) \equiv (\Delta Q_x, \Delta Q_y,\ \Delta Q_z,\ \Delta\omega)$. Expressions for M_{ij} and R_0 are presented in Appendix A and can be shown to be identical to those given by Cooper and Nathans for the beam configuration used in their paper.

C. Effect of Sample Mosiac Spread

Expression (9) gives the counting rate N for a perfect sample crystal, if the cross section $d^3\sigma/dk_2^3$ given by (6) is used. In order to include the effects of the sample mosaic spread, the expression for N in Eq. (9) should be integrated over the polar angles φ and ψ of the sample mosaic distribution $\rho(\varphi, \psi)$. That is, the total counting rate due to all mosaic grains is

$$N = \Phi_0 \frac{m_n}{\hbar}\int d\varphi \int d\psi\, \rho(\varphi, \psi) \times \iint \frac{1}{Q}\frac{d^3\sigma(\vec{Q}', \omega_{Q'})}{dk_2^3} R(\Delta\vec{Q}, \Delta\omega)\, d^3(\Delta\vec{Q})\, d(\Delta\omega), \tag{13}$$

where a wave vector $\vec{Q}'$ in a given mosaic grain specified by (ϕ, ψ) differs from the wave vector $\vec{Q} = \vec{Q}_0 + \Delta\vec{Q}$ in the mosaic grain in the center of the sample mosaic distribution by

$$\vec{Q}' = \vec{Q} + \delta\vec{Q} \tag{14}$$

and

$$\delta\vec{Q} = \psi Q_0 \hat{y} + \varphi Q_0 \hat{z}\ . \tag{15}$$

Since a numerical integration over all the variables in (13) is intractable, some approximation is necessary. We assume that the variation of the polarization vectors (and the thermal factor) over the range of $\delta\vec{Q}$ is much less than the rapid variation of R with wave vector. If we also assume the sample mosaic distribution is Gaussian in the two variables φ and ψ, we can transform the integration over $\vec{Q}$ to one over $\vec{Q}'$ and integrate over φ and ψ analytically. The result is

$$N \approx \frac{m_n}{\hbar}\Phi_0 \frac{1}{Q_0}\iint \frac{d^3\sigma(\vec{Q}', \omega_{Q'})}{dk_2^3} \times R'(\Delta\vec{Q}', \Delta\omega)\, d^3(\Delta\vec{Q}')\, d(\Delta\omega), \tag{16}$$

where

$$R' = R_0' \exp(-\tfrac{1}{2}\textstyle\sum_{ij} M_{ij}' X_i' X_j') \tag{17}$$

and

$$(X_1',\ X_2',\ X_3',\ X_4') \equiv (\Delta Q_x',\ \Delta Q_y',\ \Delta Q_z',\ \omega - \omega_{Q'}).$$

The matrix elements M_{ij}' of this "effective" resolution function are

$$M_{ij}' = M_{ij} - M_{i2}M_{j2}/(1/\eta_{SH}^2 Q_0^2 + M_{22}) \quad (\text{for } i, j = 1,\ 2,\ 4) \tag{18a}$$

and

$$M_{33}' = M_{33} - M_{33}^2/(1/\eta_{SV}^2 Q_0^2 + M_{33}), \tag{18b}$$

and the coefficient R_0' is given by

$$R_0' = R_0 (\eta_{SH}\eta_{SV})^{-1}[(1/\eta_{SV}^2 + Q_0^2 M_{33})(1/\eta_{SH}^2 + Q_0^2 M_{22})]^{-1/2}, \tag{19}$$

where η_{SH} and η_{SV} are the sample mosaic spread parameters in the horizontal and vertical planes. We note that M_{ij}' is less than M_{ij} and thus R' is more extended in ω, $\vec{Q}$ space than R.

D. Effect of the Monitor

In our experiment counts were registered per unit monitor count and thus comparison of theory and experiment requires that we should divide N by the monitor counting rate M. If the monitor is a "$1/v$" detector this procedure would appear to cancel the $1/k_1$ factor in the one-phonon cross section. However, since the monitor counts neutrons of all orders scattered by the monochromator, this may not be a good approximation if one is interested in line shapes for scans which involve a large variation of k_1.

The monitor counting rate is

$$M = M_1 + M_2 + M_3 \cdots, \tag{20}$$

where the indices refer to first-, second-, third-order neutrons. For example,

$$M_1 = \int \Phi_0(\vec{k}_1) P_M(\vec{\Delta}) \epsilon_M(\vec{k}_1)\, d^3k_1, \tag{21}$$

where $\vec{k}_1$ is the first-order wave vector and $\epsilon_M(\vec{k}_1)$ is the efficiency of the monitor for neutrons of this energy.

Expressions for M_1, M_2, M_3, etc. can be obtained if the incident flux distribution is presumed to be Maxwellian and if P_M is assumed to vary as k_1^{-3}. However, the dips in the monochromator reflectivity caused by competing reflections and the uncertainty in the variation of P_M with k_1 make this method inadequate in practice. An independent measurement of the order contamination in the beam, say, by the use of gold foils of various thickness (Stedman[9]), is probably a better way to correct for the monitor counting rate. This can be done simply by writing

$$\begin{aligned} M &= M_1\left(1 + \frac{M_2 + M_3 \cdots}{M_1}\right) \\ &= M_1[1 + f(k_1)] \end{aligned} \tag{22}$$

where $f(k_1) = n_1(2k_1) + n_1(3k_1) \ldots / n_1(k_1)$ for a $1/v$ monitor. n_1 is the neutron density (neutron/cm^3) in the incident beam for each of the orders of reflection. In our case $f(k_1)$ varies from about 0.3 at $k_1 = 3$ Å^{-1} to near zero at $k_1 = 5$ Å^{-1}. Thus, after making the correction $(1+f)$ to the calculation of N/M, the $1/k_1$ factor in the cross section can be cancelled.

III. MEASUREMENTS OF THE INSTRUMENTAL PARAMETERS

A direct measurement of $R'(\Delta\vec{Q}', \Delta\omega)$ at $\omega_0 = 0$, $\vec{Q}_0 = \vec{G}$ can be made by scanning through a Bragg point.[4,6,7] If one assumes that the resolution ellipsoids do not change appreciably in moving off the $\omega_0 = 0$ plane to a nearby point $(\vec{Q}_0 = \vec{G} + \vec{q}_0)$ in $\vec{Q}$ space, the analytical expression for R' can be replaced by a matrix of experimental values of R'.

Our point of view has been to obtain the instrumental parameters by independent measurements, and then to compare the resulting 50% ellipsoids with those obtained experimentally at a Bragg point. With certain adjustments (as explained below) we have found that the orientation and magnitudes of the principal axes of the calculated ellipsoids agree with the experimental ones to within about 15%. This appears to be within the accuracy to which the resolution function can be measured at a Bragg point. There are several difficulties which arise in the measurement of R'. The most important of these is simply the accuracy to which the spectrometer can be set reproducibly. A fairly small error (~0.01°) in the crystal angle can result in a 10% error in the location of a given point on the ellipsoid for the fine resolution used in some of our experiments. Another difficulty is that there will generally be large secondary extinction in the sample crystal at the Bragg position. Consequently, the effective neutron center of the sample can be appreciably removed from its geometric center. This results in the more pronounced use of neutrons emerging from one side of the incident collimator than the other, and likewise an acceptance function of the analyzing collimator which is not characteristic of the entire collimator. To some extent this can be circumvented by the use of narrow soller collimators on both incident and scattered beams such that all slot widths are identical and very small in comparison to the sample size. Nevertheless, the effective mosaic spread of the sample at the Bragg position may not be characteristic of the sample as a whole (which is appropriate for phonon resonances). Likewise, it is not generally appropriate to replace the sample on which the experiment is to be done by, say, a Si or Ge crystal since the extinction will be different (generally more severe) and the sample size and orientation with respect to the incident beam will, in general, not be the same as for the sample in which the phonons are to be measured. We have found that the secondary extinction is *not* severe in our sample (6 cm in diameter × 12 cm in length) as a result of the rather low atomic density. This was established by observing that the sample transmission changed only by about 2% when oriented at the (200) and (220) Bragg positions. Consequently, the difficulties due to extinction are not thought to be so important in our case.

We have found in our measurements of the resolution function at the (200) Bragg point in Na that the individual scans at constant $\vec{Q}$ are quite Gaussian near $\Delta Q = 0$, but become asymmetric at larger ΔQ. The reason for this has not been definitely established. We have also found that the effective mosaic parameter for the monochromator varies slightly with wave vector. The possible reasons for this are apparent. Namely, a progressively larger volume of the monochromator is used as the wave vector is increased, and if its mosaic structure is not homogeneous, the effective η_M may change. Also, the effect of secondary extinction on the effective width of the monochromator changes with wave vector.

In view of these difficulties, certain cross checks on the measurement of the instrumental parameters are desirable. We have found that it is necessary to make certain compromises in establishing a "good set" of parameters.

The collimator parameters α_1, β_1, α_2, and β_2 (see Appendix A for definitions) may be straightforwardly obtained. α_2 and β_2 may be found from the dimensions of the analyzer collimator while α_1 and β_1 may may be inferred from the widths of curves obtained when the analyzer collimator is rocked through the incident beam. We have found that the convolution of the two collimators is very Gaussian in both the vertical and horizontal planes.

Our initial intention was to measure η_M, α_1, and η_S by recording a large number of Bragg reflections from the Na sample and fitting the widths of the curves so obtained to the expression

$$W = 2(2\ln 2)^{1/2}[(\eta_M q_S/q_M)^2 + \eta_{SH}^2 + \alpha_1^2(1+\epsilon_S q_S/\epsilon_M q_M)^2]^{1/2}, \tag{23}$$

where W is the full width at half-maximum of a rocking curve, $q_S = \tan\theta_S$, $q_M = \tan\theta_M$, and ϵ_M and ϵ_S are defined in Appendix A. These measurements were performed with no collimation of the scattered beam, the analyzer removed, and the detector placed to accept nearly all scattered neutrons. An example of this procedure is shown in Fig. 2. We found that rocking curves for equivalent Bragg reflections had slightly different widths, an effect which was traced to the inhomogeneity of the incident beam and the asymmetric orientation of the Na sample, the axis of the cylindrical crystal being 15° off the vertical (110) axis. The inhomogeneity of the beam appears to be due to the bending process of the monochromator which has been used to increase η_M and hence the flux of incident neutrons. In addition to this difficulty, some Bragg reflections were found to be asymmetric, an effect which has not been completely explained. However, the fitting procedure outlined above (using values of W determined from the second moments of the rocking curves) gives the same value of α_1 as was determined from the convolution of the collimators.

Conversely, both η_M and η_S vary by as much as 15% when poorly fitted values of W are rejected and the fit repeated. Our calculations indicate that sample mosaic does not have a large effect on the widths of the phonon resonances which we have measured and thus the uncertainty in the value of η_S described above is not a serious problem. Conversely, an accurate value of η_M is required since the focusing of the spectrometer depends to a significant extent on this parameter. A rocking curve of a large Ge (331) crystal against the monochromator (in the parallel position) gave a mosaic width for the latter which is close to the average value obtained from the fitting procedure described above. For lack of a better choice we have therefore used this value.

A rocking curve of the Ge crystal against our analy-

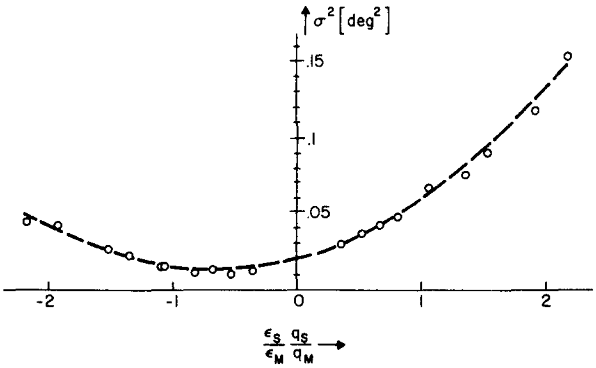


FIG. 2. Plot of the squared standard deviation σ^2 ($= W^2/5.57$) of Na rocking curves as a function of Bragg angle θ_S written in terms of the variable $\epsilon_S \tan\theta_S/\epsilon_M \tan\theta_M$. The dashed curve is the least-squares-fitted function given by Eq. (23). The parameters of this fitted curve are $\eta_M = 0.095$, $\eta_S = 0.098$, $\alpha_1 = 0.10$ deg.

zer crystal showed that the reflectivity of the latter could be closely approximated by the sum of two Gaussian curves. This phenomenon was found to be due to surface strain in the analyzer crystal and was removed before some of our later experiments by etching the analyzer crystal in nitric acid for 1.5 h.

It is perhaps worthwhile to mention a problem which led to a severe discrepancy between some of our earlier measurements and the calculations. The finer analyzer collimator which we used was constructed, for reasons of mechanical stability, from cadmium-plated copper sheets. The uneven distribution of the cadmium plating produced a collimator whose transmission function was composed of a narrow Gaussian curve (corresponding to the soller slits) superposed on a broad curve (corresponding to the collimator as a whole). The effect is not easy to observe unless the faulty collimator can be rocked against a finely collimated source and was not, for example, apparent in our measurement of the convolution of the monochromator and analyzer collimators.

We have described in some detail the methods we have used to determine the instrumental parameters in order to emphasize some of the problems which are likely to arise in any detailed analysis of the effects of resolution in neutron spectrometry. The difficulties inherent in the determination of, for example, phonon lifetimes should be apparent from the foregoing discussion. The instrumental parameters which we finally decided to use are given in Table I.

IV. DESCRIPTION OF THE COMPUTER PROGRAM

Since Eq. (16) can only be integrated analytically in very simple cases, it is generally necessary to

TABLE I. Instrumental parameters .

		Weight	Degrees
Monochromator mosaic spread η_M		1.0	0.095
Analyzer mosaic spread η_A	etched	1.0	0.032
	unetched	0.8	0.032
		0.2	0.148
Sample mosaic spread $\eta_{SH}=\eta_{SV}$		1.0	0.098
Incident collimator	horizontal α_1	1.0	0.10
	vertical β_1	1.0	0.36
Analyzer collimator I	horizontal α_2	1.0	0.245
	vertical β_2	1.0	0.490
Analyzer collimator II	horizontal α_2	1.0	0.490
	vertical β_2	1.0	0.981

perform the integration numerically. The integration is over three variables since the delta function of Eq. (6) and the integration over $\Delta\omega$ lead to the replacement of X_4 by $(\omega_{Q'}-\omega_0)$. This is equivalent to assuming that the phonon lifetime is infinitely long. To include the effect of lifetime, the delta function of Eq. (6) should be replaced by some normalized function $F(\omega-\omega_{Q'})$ which describes the natural linewidth of the resonance. However, if we can assume that the phonon lifetime does not change appreciably inside the resolution volume in $\vec{Q}$ space, the counting rate $N(\omega_0)$ can be written as

$$N(\omega_0)=\int F(\omega-\omega_0)N_6(\omega)\,d\omega\,, \tag{24}$$

where $N_6(\omega_0)$ is the counting rate obtained for a phonon of infinite lifetime.

Our program computes a quantity which is proportional to the counting rate given for phonon creation by Eq. (16). We have made estimates of the constants required in calculating the *absolute* counting rate from the output given by our program and have generally found agreement with experiment to within about 50%. The *relative* intensities of phonons is predicted excellently.

To do the integration of Eq. (16) we have expanded both the cross section and the resolution function as second-order polynomials in the variables X_i' within suitably chosen volumes in $\vec{Q}$ space and have performed the resulting integrations exactly. A rectangular prism is constructed with sides parallel to X_1', X_2', and X_3'. The size of the prism is chosen such that the resolution function has fallen to a value smaller than $\exp(-\frac{1}{2}p_{max})$ of its value at $X_i=0$ on all boundaries of the prism. p_{max} is determined by an input parameter, and a value $p_{max}=8$ has been found adequate. The rectangular prism is divided into an array of approximately cubic *cells* and the cross section is expanded within each of these about the center point. A division of the prism into about 60 cells has been found suitable for the cases which we have considered. The expansion of the quantity

$$E=\frac{n_{Q's}+1}{\omega_{Q's}}\,(\vec{Q}\cdot\vec{\epsilon}_{Q's})$$

which we have used is given in Appendix B.

Since the resolution function R' may vary rapidly within a particular cell, we expand it within a set of *subcells*. The subcell dimensions (for each cell) are decided within the program in such a way that the variation of R' over each subcell allows a second-order expansion of R' whose accuracy is consistent with ignoring contributions to the integral which fall outside the integration prism.

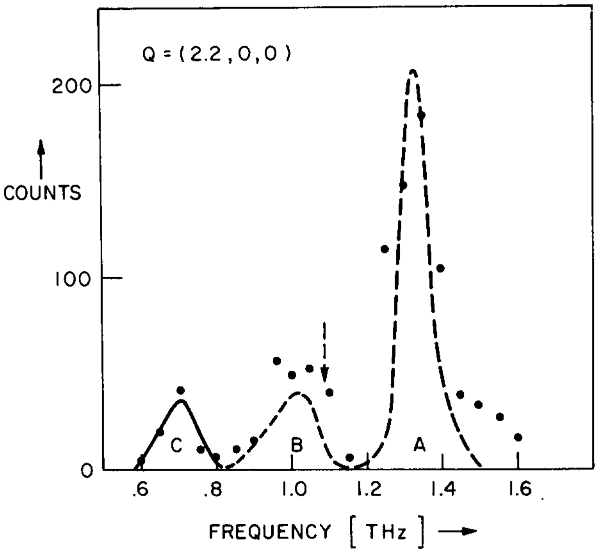
As the program is now written, the phonon frequencies and eigenvectors are computed at the center of each cell using a force-constant model. A subroutine which computes the dynamical matrix using any other appropriate model can easily be incorporated. The necessary derivatives of the frequencies and eigenvectors required by the expansions given in Appendix B are calculated using a perturbation scheme similar to that given by Gilat and Raubenheimer.[11]

The expansion scheme which we have used has the advantage of a substantial reduction in computer time since the frequencies and eigenvectors need to be calculated only at a relatively small number of points in $\vec{Q}$ space. Although the algebra involved in setting up these expansions has been rather tedious, the computational time involved in the integration of Eq. (16) has been reduced to about 1 sec for each value of ω_0 on the CDC 6600. A detailed description and listing of the program can be obtained on request.[12]

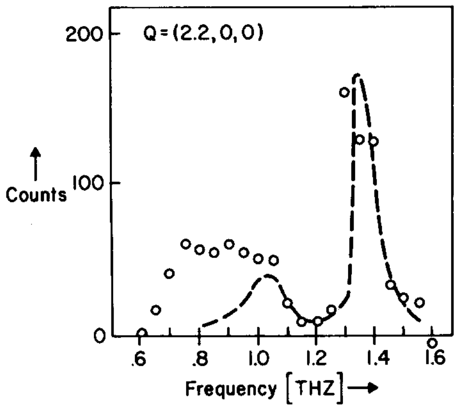
V. COMPARISON WITH EXPERIMENT

In order to illustrate some of the effects which we have observed that are directly attributable to the resolution of the spectrometer we consider examples of phonons propagating in two symmetry directions, ⟨100⟩ and ⟨111⟩, of Na. In all cases, the experiments were performed at 78 °K and the scattering plane was aligned with the (110) plane of the Na crystal. Furthermore, the beam path and the value of k_2^0 (held fixed) was always chosen in order to obtain focused resonances. The criteria used to determine focusing have been discussed by, for example, Stedman[5] who also presents an approximate method for determining the expected resolution width of a resonance. It is of interest to compare the latter width, which is based on the assumption of a planar dispersion surface over which the cross section is constant, and ignores the effects of vertical collimation, with that deduced from our full resolution calculation. Discrepancies as large as 500% have been found in some cases. In all cases we have normalized our calculated resonances to give approximately the same scattered intensity as was observed experimentally. The single crystal used in these experiments is cylindrical in shape (6 cm in diameter and 12 cm long) with a ⟨110⟩ direction 15° off the cylinder axis. The sample was prepared using a technique similar to that described by Bowers *et al.*[13]

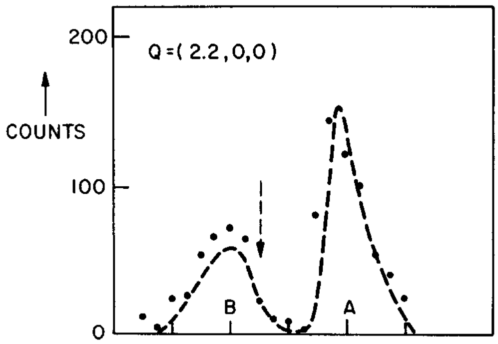
The first case we consider is that of measurements made with the neutron-scattering vector $\vec{Q}=(2.2, 0, 0)$. Since $\vec{q}$ is parallel to $\vec{Q}$ in this case we would expect to observe only the longitudinal phonon in the ⟨100⟩ direction. This phonon is found to be asymmetric as shown in Fig. 3(a). In addition to this longitudinal phonon (labelled A) we observe two additional resonances (B and C) at frequencies below this longitudinal mode. One of these (labelled C) is fairly easy to identify: It occurs at a frequency which is roughly half that of the longitudinal resonance and is due to second-order neutrons. That is, we are counting neutrons which have incident and final wave vectors which are twice the nominal values and scattering from the longitudinal phonon of wave vector $\vec{Q}=(4.4, 0, 0)$. Since the dispersion relations are fairly linear over this range of $\vec{q}$ in the ⟨100⟩ direction, the phonon frequency at $\vec{Q}=(4.4, 0, 0)$ is almost double that at $\vec{Q}=(2.2, 0, 0)$. The other "forbidden" resonance, B, is not so straightforwardly explained. In our early measurements, these two resonances, B and C,were not so well separated as shown in Fig. 3(b). However, upon improving the quality of our analyzer system, we are now convinced that there are, in general, two parts to this "extra" (100) intensity. The dashed line in Figs. 3(a) and 3(b) is the result of our resolution calculation and is seen to account for the data fairly accurately. A similar set of data for the same wave vector using more relaxed analyzer collimation (II in Table I) is shown in Fig. 3(c). The extra intensity B is seen to be larger (relative to A).



(a)



(b)



(c)

FIG. 3. (a). Constant-$\vec{Q}$ scan at (2.2, 0, 0) showing three peaks. A is the longitudinal mode, B is due to off-axis transverse phonons, and C is due to second-order neutrons scattering off the longitudinal phonon at $\vec{Q}=(4.4, 0, 0)$. Analyzer collimater I and the etched-analyzer crystal were used. A constant background level of about 100 counts was subtracted from the data. (b) Same scan as (a) before improving the analyzer resolution. (c) Same scan as (a) using the more relaxed analyzer collimator II (see Table I). The nominal scattered wave vector $k_2^0=3.1$ Å^{-1}. The beam path is given by $\epsilon_M=+1$, $\epsilon_S=+1$, and $\epsilon_A=-1$. The arrow indicates the position of the $T_{\parallel}$ resonance on the ⟨100⟩ axis at $q=0.2$.

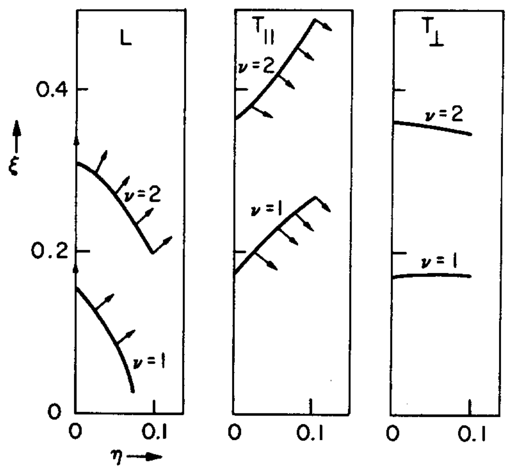


FIG. 4. The variation of the frequencies and the polarization vectors of the three branches of the dispersion surface close to the ⟨100⟩ axis in the (011) plane. The phonon wave vector is $q=(\epsilon, \eta, \eta)$ in units of $2\pi/a$.

Our resolution calculation shows that this extra resonance B comes from one of the "transverse" modes. Because Na metal is highly anisotropic elastically, the phonon frequencies change very rapidly in moving off the ⟨100⟩ direction. There is a correspondingly rapid variation of the polarization vectors. This variation is shown in Fig. 4 in which we have plotted the constant frequency surfaces and polarization vectors in the (011) plane close to the ⟨100⟩ axes.[14] The frequency of the one transverse mode labelled $T_\perp$ in Fig. 4 does not change very rapidly in moving off the ⟨100⟩ direction and its polarization remains perpendicular to (011) plane. However, the frequency and polarization vector of the other "transverse" mode $T_\parallel$ change rapidly. The frequency of this branch has a local maximum on the (100) line. The resolution volume of the spectrometer contains part of the transverse dispersion surface for which $\vec{\epsilon}$ and $\vec{Q}$ are not perpendicular and which therefore contributes to the scattered intensity. Further, since the frequency of this contributing part of the dispersion surface is lower than that on the ⟨100⟩ axis, we observe intensity which does not correspond to the frequency of the transverse mode on the ⟨100⟩ axis. This is shown schematically in Fig. 5. The resolution ellipsoids in both the vertical ($\Delta\omega$ vs ΔQ_z) and horizontal ($\Delta\omega$ vs ΔQ_y) planes are shown. In our case the resolution ellipsoid is more extended in the vertical (ΔQ_z) direction than in the horizontal plane. However, its extent in both planes is important. As the ellipsoid is moved down the ω axis (at constant Q) the intensity reaches a maximum at a frequency ω_2 which is less than the on-axis frequency ω_1. The density of the curve representing the dispersion surface is meant to show the increasing importance of the off-axis phonons resulting from the increase in the polarization factor $(\vec{Q}\cdot\vec{\epsilon})^2$. Copley[15] has explained the extra (100) intensity he has observed in Rb in a semiquantitative manner analogous to the arguments given here. In his experiments the vertical collimation was much more relaxed than in this case.

We see from a comparison of Figs. 3(a) and 3(c) that the "forbidden" intensity is not greatly affected by decreasing the analyzer collimation by a factor of 2, and it would thus appear that unreasonably tight collimation would be required if the effect was to be removed to a level of statistical insignificance. A comparison of these two figures also shows that the main longitudinal resonance is more asymmetric for the larger analyzer collimation. This effect can be explained qualitatively. The frequency of the on-axis L phonon is a local minimum. As the nominal frequency which the spectrometer is set to observe is increased, the first part of the dispersion surface to contribute to the scattered intensity is on the ⟨100⟩ axis. The intensity of scattering thus rises rapidly. As the nominal spectrometer frequency is further increased, a larger area of the dispersion surface

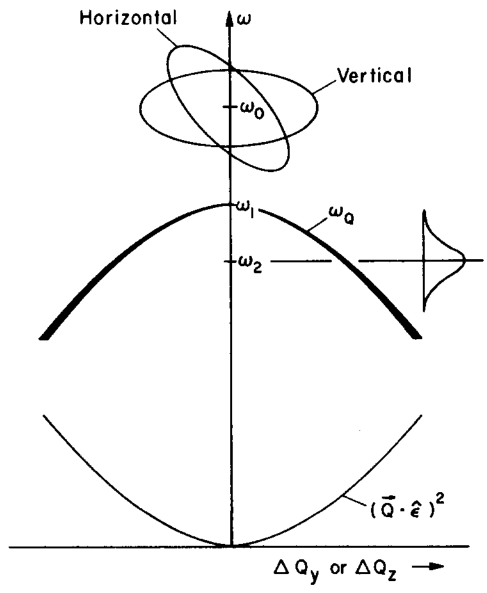


FIG. 5. A schematic diagram showing the variation of the frequency ω_Q of the transverse branch $T_\parallel$ as the wave vector moves off the ⟨100⟩ axis in either the y (horizontal) or z (vertical) directions. The density of the line is meant to indicate the relative importance of the off-axis points on the dispersion surface due to the $(\vec{Q}\cdot\vec{\epsilon})^2$ factor. The peak resulting from scanning the resolution ellipsoid along the ω axis occurs at ω_2 as shown.

contributes to the scattering. Since the dispersion surface is curved upward, scattered intensity persists to higher frequencies than if it were flat giving rise to a "tail" on the longitudinal resonance. The effect is stronger for a larger resolution volume. This asymmetry is primarily an effect of an anisotropic dispersion surface. Clearly, such effects will be most prevalent at low q where the curvature of the dispersion surface is large and will make even the determination of accurate phonon frequencies and hence elastic constants difficult.

We have found that the width of the longitudinal resonance is rather sensitive to the spectrometer parameters, and thus, the agreement between our calculations and the experimental data is *a posteriori* evidence for our choice of resolution parameters. Although the calculated curves were obtained using a force-constant model, the results are, in this case, not particularly model dependent. We obtained, for example, very similar curves from models of the dispersion relation based upon the elastic constants measured by Martinsen[16] and those deduced from our neutron data. The predicted width of the longitudinal resonance at $\vec{Q} = (2.2,\ 0\ \ 0)$ using a planar dispersion surface is $0.08T$ Hz, whereas the actual width is about $0.15T$ Hz. Similar results are obtained for the data shown in Fig. 6 for phonon at $\vec{Q} = (2.5,\ 0,\ 0)$. Thus, it would appear even in this case, that a detailed resolution calculation would be required if any reasonable assessment of the phonon lifetime is to be made.

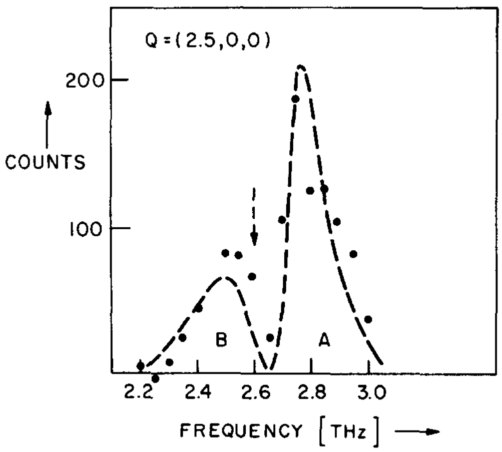


FIG. 6. The variation of the scattered intensity as a function of frequency for $\vec{Q} = (2.5,\ 0,\ 0)$ shown here is similar to that for $\vec{Q} = (2.2,\ 0,\ 0)$. The dashed arrow indicates the position of the $T_{\parallel}$ frequency on the $\langle 100\rangle$ axis. The beam path and the scattered wave vector $\vec{k}_2$ are the same as in Fig. 3.

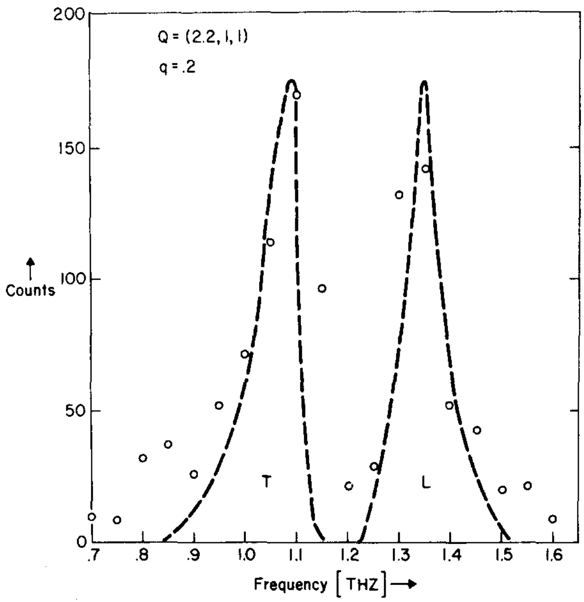


FIG. 7. A comparison of a measurement of both the (100) L and $T_{\parallel}$ phonons of wave vector $\vec{q} = 0.2$ as measured at $\vec{Q} = (2.2,\ 1,\ 1)$ with the resolution calculation (dashed line) showing the strong effects of the curvature of the dispersion surface. In this case $k_2^0 = 3.4$ Å^{-1}, $\epsilon_M = +1$, $\epsilon_S = -1$, and $\epsilon_A = +1$.

The importance of the curvature of the dispersion surface in determining the shape of the observed resonance is shown in Fig. 7 where both the longitudinal and transverse (100) phonons at $\vec{q} = 0.2$ are shown as measured at $\vec{Q} = (2.2,\ 1,\ 1)$. The asymmetry is due to the fact that the $\langle 100\rangle$ axis is a local maximum for the transverse phonon frequency and a local minimum for the longitudinal phonon frequency. Thus, asymmetry seems to be a rather general result of the spectrometer resolution volume intersecting a curved part of the dispersion surface.

As a further example of how the resolution function of the spectrometer obscures the natural phonon line shape and how much broader a resonance can be experimentally than that calculated from a planar approximation of the dispersion surface, we show in Figs. 8(a) and 8(b) (111) transverse phonons of wave vector $\vec{q} = 0.2\sqrt{3}$ and $0.3\sqrt{3}$. The width $\Delta\epsilon$ is calculated using a planar model for the dispersion surface.

The explanation of why these resonances are so broad can be obtained from Figs. 9(a) and 9(b) in which we show how the frequency of the two "transverse" modes varies as the wave vector moves off the $\langle 111\rangle$ axis. As the resolution volume is scanned along the ω axis, we obtain scattered intensity from both branches (labelled T_1

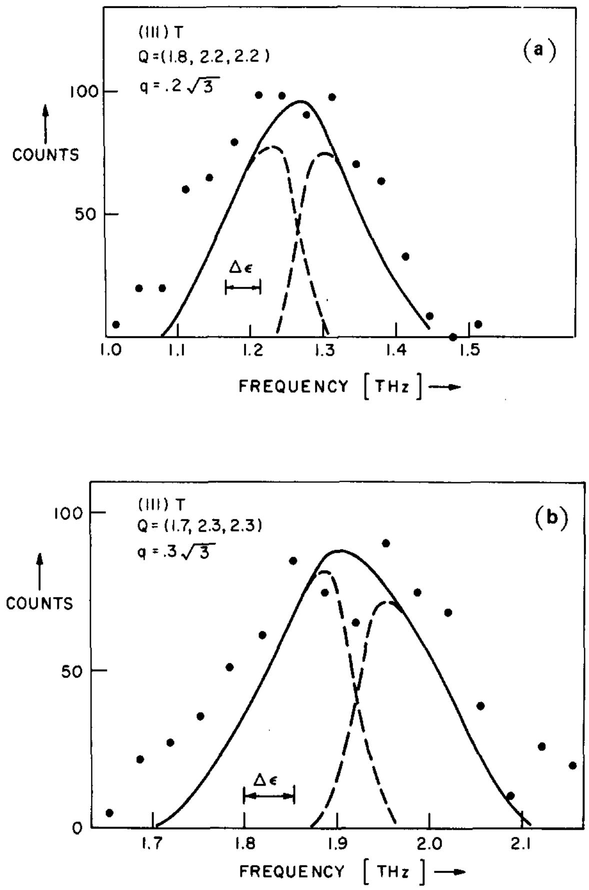


FIG. 8. (a) A comparison of the resolution calculation with a measurement of the (111) T phonon at $\vec{q}=0.2\sqrt{3}$. The dashed curves show the contributions to the total (solid curve) predicted intensity from the lower and upper "transverse" branches separately. (b) Same as (a) for the phonon of wave vector $\vec{q}=0.3\sqrt{3}$. $\Delta\epsilon$ is the predicted width of the resonance using a planar model of the dispersion surface. The nominal scattered wave vector $\vec{k}_2^0=3.1$ Å^{-1}. Beam path: $\epsilon_M=+1$, $\epsilon_S=-1$, and $\epsilon_A=+1$.

and T_2) as ω_0 approaches ω_1, the cross-over point. If the resolution ellipsoid is oriented nearly parallel to one of the branches (in one of the ω vs $\Delta q_\perp$ planes), the recorded intensity will be the sum of a broad peak and a narrow peak, both centered approximately at ω_1 as shown in Fig. 8(a). If the resolution ellipsoid is oriented as shown in Fig. 8(b), the recorded resonance will be the sum of two peaks, one displaced up and one displaced down in frequency from the on-axis frequency ω_1. Depending on the extent of the resolution function and the angle β [in Figs. 9(a) and 9(b)] this may result in a split peak. This is the case considered by Cowley and Pant[17] in their discussion of the split (111)T phonons in calcite in which they ignore the energy-momentum correlations of the spectrometer resolution. This, in general, cannot be done. There is a wide variety of resonance shapes that can be expected for the (111)T phonons depending on the orientation of the resolution ellipsoid with respect to the dispersion surface. We have also noted that if the spectrometer is not absolutely aligned on the ⟨111⟩ axis rather skewed resonances will occur. The fact that one would expect to observe broadened or split (111)T resonances in cubic materials was first pointed out by Raunio, Almquist, and Stedman[18] in their paper on the phonons in NaCl.

The predicted width of the recorded data is very sensitive to the instrumental parameters. Although the data for the $\vec{q}=0.3\sqrt{3}$ phonon shows a splitting, the calculation indicates that the two parts have coalesced together. It is apparent that the split nature of the data would be predicted exactly with a small change of the parameters. The fact that the experimental data are somewhat broader

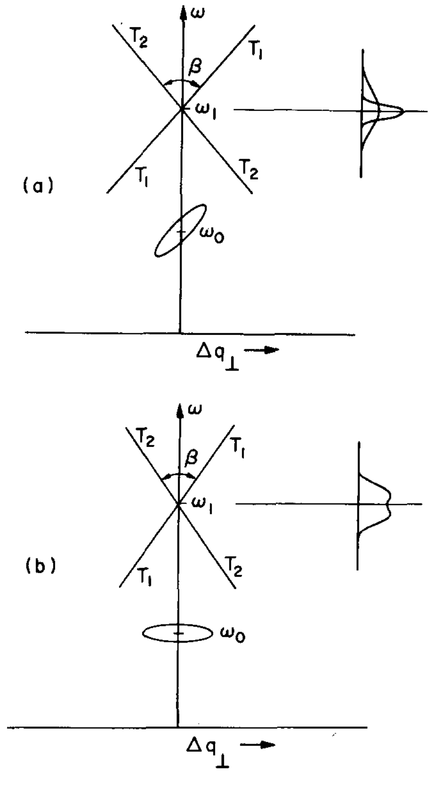


FIG. 9. Schematic diagram of the variation of the frequency of the transverse modes as the wave vector moves off the (111) axis. A variety of peak shapes can be expected depending on whether the resolution ellipsoid is aligned with one of the branches, say T_1 as shown in (a), or if it is oriented symmetrically with respect to both branches as shown in (b).

than the calculation predicts is probably due to the transparency of the analyzer collimator used in these scans as mentioned in Sec. III.

VI. CONCLUSIONS

Because sodium metal is highly anisotropic elastically, and the triple-axis neutron spectrometer samples a finite volume of $\vec{k}$ space, asymmetric, broadened, split, and "forbidden" phonon resonances are frequently observed. We have shown that a detailed resolution calculation can account for these unusual features. The phonons propagating along the $\langle 100 \rangle$ and $\langle 111 \rangle$ directions present the most serious difficulties. It is clear now that most of the anomalous features of the phonon resonances observed in Na (and probably also in the other alkali metals) are resolution effects. In order to obtain highly accurate dispersion curves (and phonon lifetimes) in a material as anisotropic as Na these resolution effects must be taken into account.

ACKNOWLEDGMENTS

We would like to thank Studsvik and our colleagues there for the hospitality afforded us during our stay in Sweden. We would especially like to acknowledge the benefit we have received from many useful discussions with Dr. R. Stedman. One of us (RP) wishes to thank the Royal Society, England for a Fellowship to support his work at Studsvik, and SW would like to thank the Swedish government for providing him with a grant to work at Studsvik. We would also like to thank Dr. J. R. D. Copley for communicating some of his results on Rb to us before publication and for several useful discussions with one of us (SW).

APPENDIX A: M_{ij} AND R_0

We give here formulas for M_{ij} and R_0 which can be shown to be equivalent to those given by Cooper and Nathans for the "W" beam configuration. The Gaussian expressions for $P_M(\vec{\Delta})$ and $P_A(\vec{\delta})$ are

TABLE II. Definitions of W_i and H_i [values of Δ_i and δ_i for which P_M and P_A equal $p_M \exp(-\frac{1}{2})$ and $p_A \exp(-\frac{1}{2})$, respectively].

Mosaic spread of monochromator, η_M	$W_1 = k_1^0 \eta_M \cot\theta_M$
Mosaic spread of analyzer, η_A	$H_1 = k_2^0 \eta_A \cot\theta_A$
Horizontal collimation between monochromator and sample, α_1	$W_2 = k_1^0 \alpha_1 \csc\theta_M$
Vertical collimation between monochromator and sample, β_1	$W_3 = k_1^0 \beta_1$
Horizontal collimation between sample and analyzer, α_2	$H_2 = k_2^0 \alpha_2 \csc\theta_A$
Vertical collimation between sample and analyzer, β_2	$H_3 = k_2^0 \beta_2$

$$P_M(\vec{\Delta}) = p_M \exp\left(-\frac{\Delta_1^2}{2W_1^2} - \frac{\Delta_2^2}{2W_2^2} - \frac{\Delta_3^2}{2W_3^2}\right), \tag{A1}$$

$$P_A(\vec{\delta}) = p_A \exp\left(-\frac{\delta_1^2}{2H_1^2} - \frac{\delta_2^2}{2H_2^2} - \frac{\delta_3^2}{2H_3^2}\right). \tag{A2}$$

The definitions of W_i and H_i are given in Table II. p_M is the peak reflectivity of the monochromator (a number between 0 and 1). p_A is the peak reflectivity of the analyzer multiplied by the detector efficiency (again a number between 0 and 1). The matrix elements are given by

$$M_{kl} = \vec{V}_k \cdot \vec{V}_l - (\vec{V}_0 \cdot \vec{V}_k)(\vec{V}_0 \cdot \vec{V}_l)/|V_0|^2 \quad (k, l = 1,\ 2,\ 4)$$

$$M_{33} = [(\beta_2 k_2^0)^2 + (\beta_1 k_1^0)^2]^{-1}, \tag{A3}$$

where the vector $\vec{V}$ is given by

$$\vec{V}_j = (s_j,\ r_j,\ p_j,\ q_j). \tag{A4}$$

The definitions of the components of $\vec{V}_j$ in terms of the instrumental parameters are given in Table III where

$$u_1 = m\omega_0/\hbar Q_0^2 + \tfrac{1}{2}, \quad u_2 = \epsilon_S [(k_1^0/Q_0)^2 - u_1^2]^{1/2}$$

$$u_4 = m/\hbar Q_0$$

TABLE III. Definitions of s_j, r_j, p_j, and q_j.

j	$\eta_M s_j$	$\eta_A r_j$	$\alpha_1 p_j$	$\alpha_2 q_j$
0	$(T_M a + b/k_1^0)$	$(T_A A - B/k_2^0)$	b/k_1^0	B/k_2^0
1	$(T_M b - a/k_1^0)(u_1 - 1)$	$(T_A B + A/k_2^0)u_1$	$a(1-u_1)/k_1^0$	$-Au_1/k_2^0$
2	$(T_m b - a/k_1^0)u_2 - (T_M a + b/k_1^0)$	$(T_A B + A/k_2^0)u_2$	$-au_2/k_1^0 - b/k_1^0$	$-Au_2/k_2^0$
4	$(T_M b - a/k_1^0)u_4$	$(T_A B + A/k_2^0)u_4$	$-au_4/k_1^0$	$-Au_4/k_2^0$

and

$$T_M = \epsilon_M \tan\theta_M / k_1^0, \qquad T_A = \epsilon_A \tan\theta_A / k_2^0,$$

$$a = \epsilon_S \sin\phi, \qquad b = \cos\phi,$$

$$A = \epsilon_S \sin(\phi + 2\theta_S), \quad B = \cos(\phi + 2\theta_S).$$

It will be noted that the advantage of writing M_{ij} in this form is that the dependence on each of the components in the system is explicitly given. That is, all s_j depend only on η_M, all r_j depend only on η_A, etc. The beam path is defined by the parameters ϵ_M(monochromator), ϵ_S(sample), and ϵ_A(analyzer) which are ± 1 or -1 depending on whether the scattering is to the left or to the right at the monochromator, sample, or analyzer, respectively.

It should be pointed out that the rather complicated formula for the width of a phonon resonance given by Cooper and Nathans [Eq. (67-d) of Ref. 4] can be shown to be equivalent to the much simpler formula given by Stedman[5] for constant-$\vec{Q}$ scans, in which there are no cross terms in the parameters η_M, α_1, α_2, and η_A. That is, each component contributes separately to the resolution width for the case of a planar dispersion surface.

The coefficient R_0 is given by

$$R_0 = 2\pi p_M p_A \{[(1/\beta_1 k_1^0)^2 + (1/\beta_2 k_2^0)^2]\,(s_0^2 + r_0^2 + p_0^2 + q_0^2)\}^{-1/2}. \tag{A5}$$

APPENDIX B: EXPANSION OF E AND R'

The expansion which we use for $E = [(n_{Q's}+1)\omega_{Q's}^{-1}] \times (\vec{Q}\cdot\vec{\epsilon})^2$ is

$$E = E_c + \sum_{k=1}^{3} \alpha_k (X_k - X_{kc}) + \tfrac{1}{2} \sum_{k,l=1}^{3} \gamma_{kl} (X_k - X_{kc})(X_l - X_{lc}), \tag{B1}$$

where E_c is the value of E at the center of a cell and X_{kc} are the coordinates of that point. The constants α_k and γ_{kl} are given by

$$\alpha_k = A_k + B_k, \tag{B2}$$

$$A_k = \left[(\vec{Q}\cdot\vec{\epsilon})^2 \left(\omega \frac{\partial n}{\partial \omega} - n - 1\right) \frac{\partial \omega}{\partial X_k}\right] \omega^{-2}, \tag{B3}$$

$$B_k = 2\left(\frac{n+1}{\omega}\right) (\vec{Q}\cdot\epsilon) \left(\epsilon_k + \sum_j X_j \frac{\partial \epsilon_j}{2X_k}\right), \tag{B4}$$

and

$$\gamma_{kl} = A_{kl} + B_{kl}, \tag{B5}$$

$$A_{kl} = 2A_k \left[\left(\epsilon_l + \sum_j Q_j \frac{\partial \epsilon_j}{\partial X_l} \Big/ (\vec{Q}\cdot\vec{\epsilon}) - \frac{\partial \omega}{\partial X_l}\right) \omega^{-1}\right]$$

$$+ \left((\vec{Q}\cdot\vec{\epsilon})^2 \frac{\partial \omega}{\partial X_l} \frac{\partial \omega}{\partial X_k} \frac{\partial^2 n}{\partial \omega^2}\right) \omega^{-1}, \tag{B6}$$

$$B_{kl} = \left[(B_l + 2A_l)\left(\epsilon_k + \sum_j Q_j \frac{\partial \epsilon_j}{\partial X_k}\right)\right.$$

$$\left. + 2E_c \left(\frac{\partial \epsilon_k}{\partial X_l} + \frac{\partial \epsilon_l}{\partial X_k}\right)\right] (\vec{Q}\cdot\vec{\epsilon})^{-1}, \tag{B7}$$

where the subscript $(\vec{Q}', s)$ and the primes on X_k are omitted for convenience.

The expansion of R' which we use is

$$R' = R_0' \exp -\tfrac{1}{2}\left[P_s + 2 \sum_{i=1}^{3} D_i (X_i - X_{is})\right.$$

$$\left. + \sum_{ij=1}^{3} D_{ij} (X_i - X_i)(X_j - X_{js})\right], \tag{B8}$$

where the X_{is} are the coordinates of the center of the subcell and p_s is the value of p at that point. The coefficients appearing in Eq. (B8) are

$$D_i = \sum_{k=1}^{4} \left(M_{ik} X_{ks} + M_{4k} X_{ks} \frac{\partial \omega}{\partial X_i}\right), \tag{B9}$$

$$D_{ij} = M_{ij} + M_{4i} \frac{\partial \omega}{\partial X_j} + M_{4j} \frac{\partial \omega}{\partial X_i} + M_{44} \frac{\partial \omega}{\partial X_i} \frac{\partial \omega}{\partial X_j}. \tag{B10}$$

The primes on M_{ij} and X_j have been omitted for convenience.

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*This work was carried out at Studsvik, A.B. Atomenergi, Nyköping, Sweden, while the authors were guest scientists.

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